School Curriculum and Standards Authority

JUDGING STANDARDS

Reporting against the Achievement Standard

The annotated work samples in *Judging Standards* support teachers when reporting against the achievement standards; when giving assessment feedback; and when explaining the differences between one student's achievement and another's.

Grey highlighting identifies those aspects of the achievement standard addressed in the work sample. Annotations in black text refer to the assessment pointers while those in coloured text highlight additional, specific qualities evident in the work.

WORK SAMPLE: YEAR 10 SCIENCE

PERFORMANCE ASSOCIATED WITH GRADE A, REPRESENTING EXCELLENT ACHIEVEMENT

SUMMARY OF TASK:

Gene technology: Induced pluripotent stem cells

Students selected an area of gene technology to research and presented their information on a poster using text and graphics. Part of the task involved identifying positives and negatives related to this type of technology.

YEAR 10 SCIENCE ACHIEVEMENT STANDARD

By the end of Year 10, students analyse how the periodic table organises elements and use it to make predictions about the properties of elements. They explain how chemical reactions are used to produce particular products and how different factors influence the rate of reactions. They explain the concept of energy conservation and represent energy transfer and transformation within systems. They apply relationships between force, mass and acceleration to predict changes in the motion of objects. Students describe and analyse interactions and cycles within and between Earth's spheres. They evaluate the evidence for scientific theories that explain the origin of the universe and the diversity of life on Earth. They explain the processes that underpin heredity and evolution. Students analyse how the models and theories they use have developed over time and discuss the factors that prompted their review.

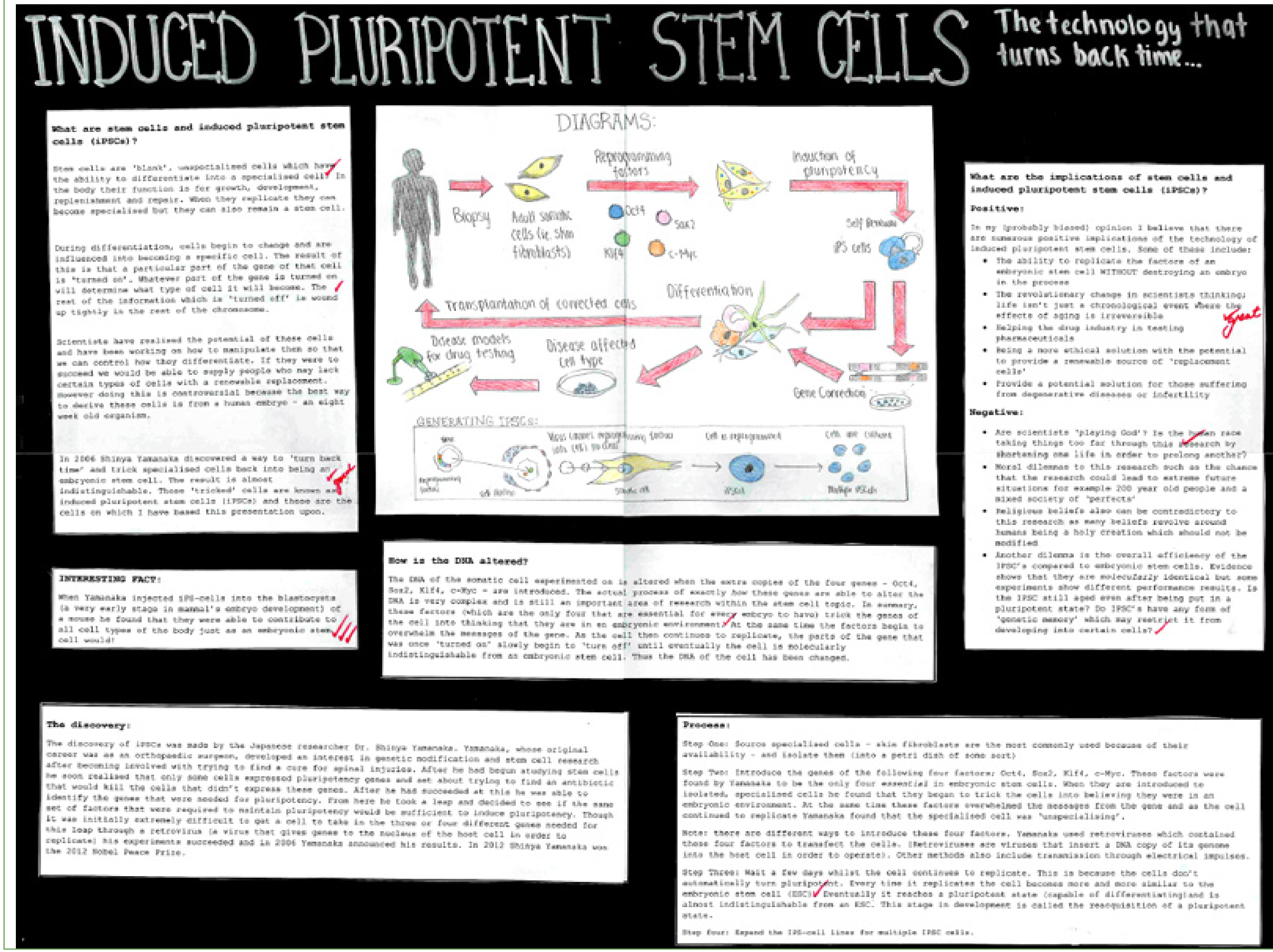
Students develop questions and hypotheses and independently design and improve appropriate methods of investigation, including field work and laboratory experimentation. They explain how they have considered reliability, safety, fairness and ethical actions in their methods and identify where digital technologies can be used to enhance the quality of data. When analysing data, selecting evidence and developing and justifying conclusions, they identify alternative explanations for findings and explain any sources of uncertainty. Students evaluate the validity and reliability of claims made in secondary sources with reference to currently held scientific views, the quality of the methodology and the evidence cited. They construct evidence-based arguments and select appropriate representations and text types to communicate science ideas for specific purposes.

KEY WORDS:

Genes, DNA, Genetic manipulation, Genetic engineering

Student achievement is reported at the end of the semester or year using the letter grades and achievement descriptors. Letter grades and achievement descriptors should not be used to assess individual pieces of work.

WORK SAMPLE



ANNOTATIONS

Presents scientific ideas and information in a logical sequence using appropriate headings and subheadings, including a range of representations where appropriate, e.g. a poster with segments of information in paragraphs and dot points, plus two detailed diagrams.

School Curriculum and Standards Authority

WORK SAMPLE

ANNOTATIONS

(This page and the following two contain close-ups of the information on the poster)

What are stem cells and induced pluripotent stem cells (iPSCs)?

Stem cells are 'blank', unspecialised cells which have the ability to differentiate into a specialised cell. In the body their function is for growth, development, replenishment and repair. When they replicate they can become specialised but they can also remain a stem cell.

During differentiation, cells begin to change and are influenced into becoming a specific cell. The result of this is that a particular part of the gene of that cell is 'turned on'. Whatever part of the gene is turned on will determine what type of cell it will become. The rest of the information which is 'turned off' is wound up tightly in the rest of the chromosome.

Scientists have realised the potential of these cells and have been working on how to manipulate them so that we can control how they differentiate. If they were to succeed we would be able to supply people who may lack certain types of cells with a renewable replacement. However doing this is controversial because the best way to derive these cells is from a human embryo - an eight week old organism.

In 2006 Shinya Yamanaka discovered a way to 'turn back time' and trick specialised cells back into being an embryonic stem cell. The result is almost indistinguishable. These 'tricked' cells are known as induced pluripotent stem cells (iPSCs) and these are the cells on which I have based this presentation upon.

INTERESTING FACT:

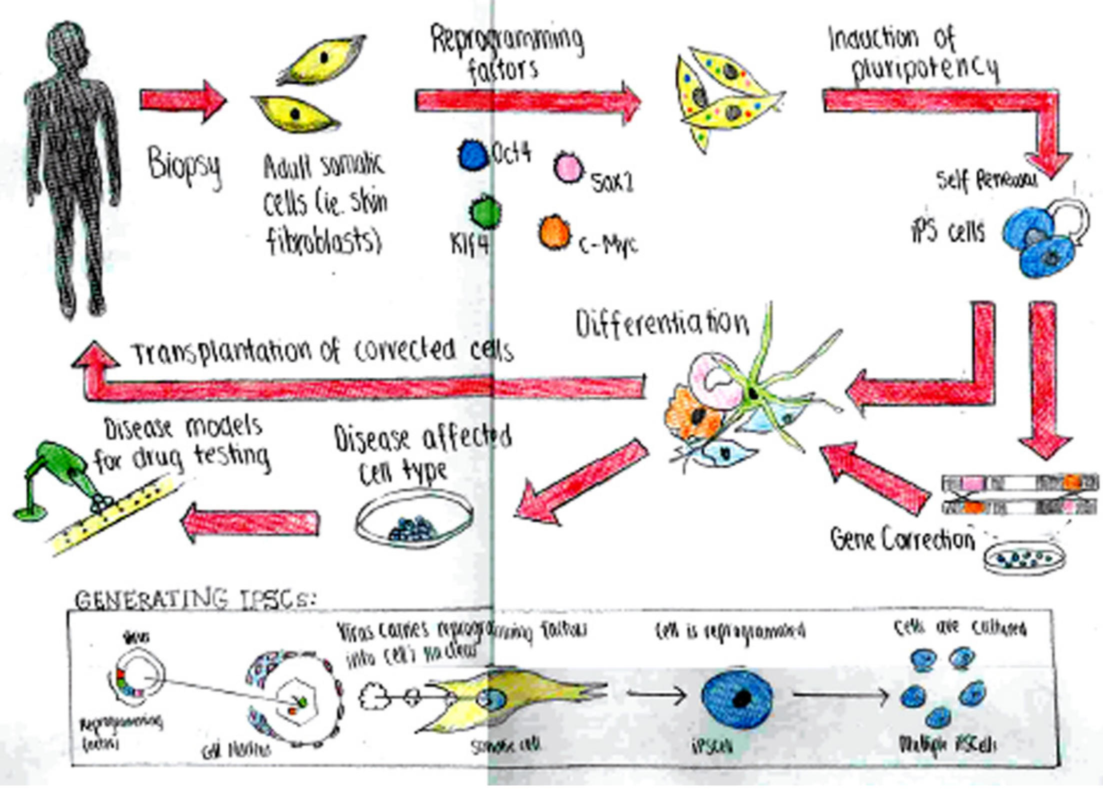
When Yamanaka injected iPS-cells into the blastocysts (a very early stage in mammal's embryo development) of a mouse he found that they were able to contribute to all cell types of the body just as an embryonic stem cell would!

Provides correct explanations of scientific theories, systems and concepts, using correct scientific terminology and supported with everyday examples, e.g. 'When Yanaka injected IPS cells into the blastocysts…'

The discovery:

The discovery of iPSCs was made by the Japanese researcher Dr. Shinya Yamanaka. Yamanaka, whose original career was as an orthopaedic surgeon, developed an interest in genetic modification and stem cell research after becoming involved with trying to find a cure for spinal injuries. After he had begun studying stem cells he soon realised that only some cells expressed pluripotency genes and set about trying to find an antibiotic that would kill the cells that didn't express these genes. After he had succeeded at this he was able to identify the genes that were needed for pluripotency. From here he took a leap and decided to see if the same set of factors that were required to maintain pluripotency would be sufficient to induce pluripotency. Though it was initially extremely difficult to get a cell to take in the three or four different genes needed for this leap through a retrovirus (a virus that gives genes to the nucleus of the host cell in order to replicate) his experiments succeeded and in 2006 Yamanaka announced his results. In 2012 Shinya Yamanaka won the 2012 Nobel Peace Prize.

DIAGRAMS:



Uses correctly labelled diagrams to enhance explanations of complex scientific models, e.g. 'Generating IPSCs'.

How is the DNA altered?

The DNA of the somatic cell experimented on is altered when the extra copies of the four genes - Oct4, Sox2, Klf4, c-Myc - are introduced. The actual process of exactly how these genes are able to alter the DNA is very complex and is still an important area of research within the stem cell topic. In summary, these factors (which are the only four that are essential for every embryo to have) trick the genes of the cell into thinking that they are in an embryonic environment. At the same time the factors begin to overwhelm the messages of the gene. As the cell then continues to replicate, the parts of the gene that was once 'turned on' slowly begin to 'turn off' until eventually the cell is molecularly indistinguishable from an embryonic stem cell. Thus the DNA of the cell has been changed.

Process:

Step One: Source specialised cells - skin fibroblasts are the most commonly used because of their availability - and isolate them (into a petri dish of some sort)

Step Two: Introduce the genes of the following four factors; Oct4, Sox2, Klf4, c-Myc. These factors were found by Yamanaka to be the only four *essential* in embryonic stem cells. When they are introduced to isolated, specialised cells he found that they began to trick the cells into believing they were in an embryonic environment. At the same time these factors overwhelmed the messages from the gene and as the cell continued to replicate Yamanaka found that the specialised cell was 'unspecialising'.

Note: there are different ways to introduce these four factors. Yamanaka used retroviruses which contained these four factors to transfect the cells. (Retroviruses are viruses that insert a DNA copy of its genome into the host cell in order to operate). Other methods also include transmission through electrical impulses.

Step Three: Wait a few days whilst the cell continues to replicate. This is because the cells don't automatically turn pluripotent. Every time it replicates the cell becomes more and more similar to the embryonic stem cell (ESC) Eventually it reaches a pluripotent state (capable of differentiating)and is almost indistinguishable from an ESC. This stage in development is called the reacquisition of a pluripotent state.

Step four: Expand the IPS-cell lines for multiple IPSC cells.

Provides correct explanations of scientific theories, systems and concepts, using correct scientific terminology and supported with everyday examples, e.g. 'Eventually it reaches a pluripotent state (capable of differentiating) and is almost indistinguishable for an ESC. This stage is called the reacquisition of a pluripotent state.'

What are the implications of stem cells and induced pluripotent stem cells (iPSCs)?

Positive:

In my (probably biased) opinion I believe that there are numerous positive implications of the technology of induced pluripotent stem cells. Some of these include:

- The ability to replicate the factors of an embryonic stem cell WITHOUT destroying an embryo in the process
- The revolutionary change in scientists thinking; life isn't just a chronological event where the effects of aging is irreversible
- Helping the drug industry in testing pharmaceuticals
- Being a more ethical solution with the potential to provide a renewable source of 'replacement cells'
- Provide a potential solution for those suffering from degenerative diseases or infertility

Great

Negative:

- Are scientists 'playing God'? Is the human race taking things too far through this research by shortening one life in order to prolong another?
- Moral dilemmas to this research such as the chance that the research could lead to extreme future situations for example 200 year old people and a mixed society of 'perfects'
- Religious beliefs also can be contradictory to this research as many beliefs revolve around humans being a holy creation which should not be modified
- Another dilemma is the overall efficiency of the IPSC's compared to embryonic stem cells. Evidence shows that they are molecularly identical but some experiments show different performance results. Is the IPSC still aged even after being put in a pluripotent state? Do IPSC's have any form of 'genetic memory' which may restrict it from developing into certain cells?

Predicts and explains changes in a system using scientific knowledge, e.g. benefits of gene technology.